Algorithms 2

Graph Colouring - Brooks' Theorem

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1. A Greedy Colouring Algorithm

1.1 Setup

Definition 1.1.1 Input:

- A graph G
- An ordering $v_1, v_2, ..., v_n$ of $V(G)$

Output:

- A legal vertex-colouring of G

- Scan the vertices according to the ordering.
- When considering v_i, assign it the least available colour.
- This always produces a legal colouring.

Question 1.1.2 How useful is this algorithm for bounding $\chi(G)$?

1.2 First Consequence

- For every graph G, define

$$\Delta(G) := \max\left\{\deg_{G(v)} : v \in V(G)\right\}$$

- The greedy algorithm implies

$$\chi(G) \leq \Delta(G) + 1$$

1.3 When The Bound Is Tight

- There are graphs with

$$\chi(G) = \Delta(G) + 1$$

- Examples:
  - Odd cycles: $\Delta = 2$ and $\chi = 3$
  - Complete graphs: $\Delta(K_r) = r - 1$ and $\chi(K_r) = r$

2. Brooks' Theorem

2.1 Statement

Theorem 2.1.1 (Brooks' Theorem) Let G be a connected graph. Then

$$\chi(G) \leq \Delta(G)$$

unless G is complete or an odd cycle.

Observation 2.1.2 Brooks' upper bound can be far from optimal in general.

2.2 Proof Strategy

- Assume $\Delta(G) \geq 3$.
- If $\Delta(G) \leq 2$, then connected G is a path, cycle, or odd cycle, so the theorem is immediate.
- Distinguish two cases:
  - There exists $v \in V(G)$ with $\deg(v) < \Delta(G)$.
  - Otherwise G is $\Delta(G)$-regular.

2.3 Case 1: A Vertex Of Smaller Degree Exists

- Choose $v \in V(G)$ with $\deg(v) < \Delta(G)$.
- Let T be a spanning tree of G rooted at v.
- Order vertices so descendants appear before ancestors and v is last.
- In this ordering, each vertex except possibly v has at most $\Delta(G) - 1$ earlier neighbours.
- Greedy colouring then yields a $\Delta(G)$-colouring.

2.4 Case 2: Regular Case

- Assume G is $\Delta(G)$-regular.
- If $\kappa(G) = 1$, then G has a cut-vertex.
- Colour components around the cut-vertex separately and permute colour names to combine them.
- Therefore we may reduce to the 2-connected case:

$$\kappa(G) \geq 2$$

2.5 Key Reduction In The 2-Connected Case

- It is enough to find vertices $x, y, z \in V(G)$ such that:
  - $xy, xz \in E(G)$
  - $yz \notin E(G)$
  - $G - \{y, z\}$ is connected
- If such vertices exist:
  - Build a spanning tree of $G - \{y, z\}$ rooted at x.
  - Use an order with x last and y, z first.
  - Since y and z are nonadjacent, they may receive the same colour.
  - Greedy colouring then gives a legal $\Delta(G)$-colouring.

2.6 Lemma Completing The Proof

Lemma 2.6.1 (Structural Lemma) Let $3 \leq k \in \mathbb{N}$ and let G be a k-regular, 2-connected, non-complete graph. Then there exist $x, y, z \in V(G)$ such that:

1. $xy, xz \in E(G)$
2. $yz \notin E(G)$
3. $G - \{y, z\}$ is connected.

2.7 Why The Lemma Holds (Outline)

- First subcase: there exists v such that $G - v$ is connected but not 2-connected.
  - Analyze the block tree of $G - v$.
  - Pick y, z in different leaf blocks, and set $x := v$.
  - Then $G - \{y, z\}$ stays connected.
- Second subcase: $\kappa(G - v) \geq 2$ for every $v \in V(G)$.
  - Since G is not complete, pick nonadjacent y, z among neighbours of some x.
  - Connectivity conditions force $G - \{y, z\}$ to remain connected.

2.8 Conclusion

Theorem 2.8.1 (Brooks' Theorem - Final Form) For every connected graph G,

$$\chi(G) \leq \Delta(G)$$

unless G is complete or an odd cycle.

- Greedy colouring gives $\Delta(G) + 1$ in general.
- The theorem shows one colour can always be saved outside the two exceptional families.

3. Edge Colouring

3.1 The Chromatic Index

Definition 3.1.1 An edge-colouring of a graph G with k colours is a function

$$\varphi : E(G) \to [k]$$

Definition 3.1.2 An edge-colouring is called proper if incident edges receive distinct colours.

Definition 3.1.3 The chromatic index of G, denoted by $\chi'(G)$, is the least k such that G has a proper edge-colouring with k colours.

Definition 3.2.1 The line graph of G, denoted $L(G)$, is the graph with:

- $V(L(G)) = E(G)$
- two vertices adjacent iff the corresponding edges of G are incident.

Observation 3.2.2

$$\chi(L(G)) = \chi'(G)$$

- Also, for every graph G:

$$\Delta(G) \leq \chi'(G)$$

3.3 A Quick Bound Via Brooks

- For an edge $uv \in E(G)$, the corresponding vertex in $L(G)$ has degree

$$\deg_{L(G)}(uv) = \deg_{G(u)} + \deg_{G(v)} - 2$$

- Hence

$$\Delta(L(G)) \leq 2(\Delta(G) - 1)$$

- Applying Brooks to $L(G)$ (when not exceptional) yields

$$\chi'(G) = \chi(L(G)) \leq 2(\Delta(G) - 1)$$

4. Konig's Theorem

4.1 Bipartite Edge Colouring

Theorem 4.1.1 (Konig's Theorem) If G is bipartite, then

$$\chi'(G) = \Delta(G)$$

4.2 Proof Idea (Outline)

- Degree-regular bipartite graphs admit a perfect matching.
- Removing edges of a perfect matching from a k-regular bipartite graph gives a $(k-1)$-regular bipartite graph.
- By induction on k, every k-regular bipartite graph is k-edge-colourable.
- Any bipartite graph can be embedded into a $\Delta(G)$-regular bipartite supergraph.
- Restrict the colouring from the supergraph back to G.

5. Vizing's Theorem

5.1 Statement

Theorem 5.1.1 (Vizing's Theorem) For every simple graph G,

$$\chi'(G) \le \Delta(G) + 1$$

- Combined with the trivial lower bound:

$$\Delta(G) \le \chi'(G) \le \Delta(G) + 1$$

- So every simple graph is either Class 1 ($\chi'(G) = \Delta(G)$) or Class 2 ($\chi'(G) = \Delta(G) + 1$).

5.2 Proof Skeleton

- Proof by induction on $e(G)$.
- Pick $e = uv \in E(G)$ and set $G' := G - e$.
- By induction hypothesis, G' has a proper edge-colouring with at most $\Delta(G) + 1$ colours.
- Try to colour e when re-inserting it.

5.3 The Recolouring Idea

- Each endpoint misses at least one colour among $[\Delta(G) + 1]$.
- If there is a common missing colour at both u and v, assign it to e.
- Otherwise, use alternating two-colour chains (Kempe chains) to swap colours along a maximal chain.
- These swaps preserve properness and can free a colour at one endpoint.
- Repeat this controlled recolouring until e can be coloured.

5.4 Conclusion

Theorem 5.4.1 (Vizing's Theorem - Final Form) Every simple graph G satisfies

$$\chi'(G) \in \{\Delta(G), \Delta(G) + 1\}$$

- Determining whether $\chi'(G) = \Delta(G)$ can be difficult in general.